Recognition, retrieval and confident errors

Multi-digit addition, a blind simulation, swapping memories, and a second task

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October 2026

Recognition, retrieval and confident errors



- The first detailed talk. It follows one model, multi-digit addition, from its postulates to a theorem that two different tasks inherit.
- Everything quoted here is a Lean theorem or an exact computation checked by Lean, unless marked otherwise.

Two theories, combined

EPAM (Feigenbaum 1959)

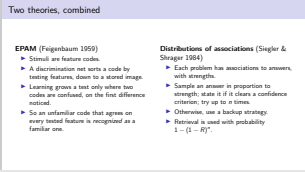
- Stimuli are feature codes.
- A discrimination net sorts a code by testing features, down to a stored image.
- Learning grows a test only where two codes are confused, on the first difference noticed.
- So an unfamiliar code that agrees on every tested feature is *recognized as* a familiar one.

Distributions of associations (Siegler & Shrager 1984)

- Each problem has associations to answers, with strengths.
- Sample an answer in proportion to strength; state it if it clears a confidence criterion; try up to n times.
- Otherwise, use a backup strategy.
- Retrieval is used with probability $1 - (1 - R)^n$.

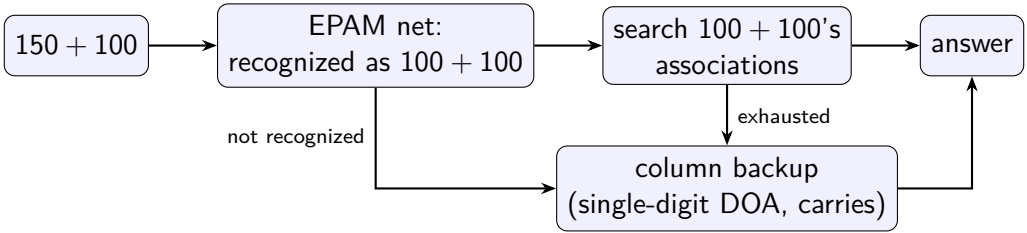
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Two theories, combined



- R is the chance that one search clears the criterion. Both theories' phenomena are already proved in the library.
- In DOA, a flat distribution raises both errors and backup use; they go together. That fact is what the next slides break.

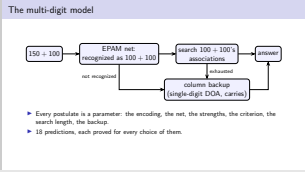
The multi-digit model



- Every postulate is a parameter: the encoding, the net, the strengths, the criterion, the search length, the backup.
- 18 predictions, each proved for every choice of them.

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The multi-digit model



- The column backup is built from single-digit DOA trials chained by carries. Carrying adds no error: a column error is a single-digit error moved to its place.
- The predictions include near misses (within 200 or 20, depending on which digits the net tests), harmless confusions (same sum, commuted problems), and no interference with practiced problems.

Confident errors

Let a familiar problem q have *sharp* associations (only its correct sum clears the criterion), and let x, with a different sum, be recognized as q. Then:

1. on q, whole-problem retrieval never errs;
2. on x, every whole-problem answer is wrong;
3. x uses the backup exactly as rarely as q does: $(1 - R_q)^n$.

In DOA alone, errors and backup use rise together. Confusion breaks the tie: a retrieved, wrong answer with no extra backup use.

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Confident errors

- This is the prediction neither theory makes alone.
- Two cautions. The model has no timing, so "fast" would need a timing model. And the errors are frequent only if retrieval on q usually succeeds (R_q near 1); if $R_q = 0$, every trial uses the backup.

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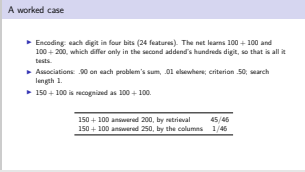
A worked case

- Encoding: each digit in four bits (24 features). The net learns $100 + 100$ and $100 + 200$, which differ only in the second addend's hundreds digit, so that is all it tests.
- Associations: .90 on each problem's sum, .01 elsewhere; criterion .50; search length 1.
- $150 + 100$ is recognized as $100 + 100$.

$150 + 100$ answered 200, by retrieval	45/46
$150 + 100$ answered 250, by the columns	1/46

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A worked case



- These numbers are for illustration, computed exactly by the Lean kernel. The structural result, confident errors, holds for every parameter choice.

Checking the formalization: a blind simulation

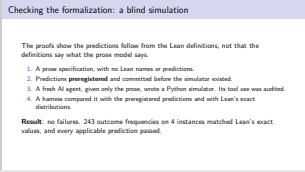
The proofs show the predictions follow from the Lean definitions, not that the definitions say what the prose model says.

1. A prose specification, with no Lean names or predictions.
2. Predictions **preregistered** and committed before the simulator existed.
3. A fresh AI agent, given only the prose, wrote a Python simulator. Its tool use was audited.
4. A harness compared it with the preregistered predictions and with Lean's exact distributions.

Result: no failures. 243 outcome frequencies on 4 instances matched Lean's exact values, and every applicable prediction passed.

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Checking the formalization: a blind simulation



- This is N-version programming applied to a theory: two independent implementations of one specification.
- Five predictions were vacuous: their hypotheses (a path testing every feature of a digit) never held in nets EPAM actually learns. True theorems that say little about real nets: a finding.
- It also found a gap (zero total strength gives a trial no mass in Lean; the natural reading is “retrieval exhausted”), and that the learning sequence depends on a choice the model leaves open: what a newly discriminated problem starts with.

Memory as a parameter

The recognition step is now an interface: memory takes a probe and returns a *distribution* over stored items (or nothing).

- EPAM's net is one memory: deterministic.
- ACT-R's declarative memory is another: base-level activation that decays, partial matching, a retrieval threshold.
- With decay 1 and temperature 1, ACT-R's retrieval probabilities are exactly rational, so Lean computes them exactly.

Each prediction now states the law of memory it needs. Swap the memory and keep every theorem whose laws still hold.

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- Other decay rates and temperatures need real exponentials; they would go in a separate package with Mathlib.

The same problems, two memories

	EPAM	ACT-R
$150 + 100$ answered 200 (the neighbour's sum)	45/46	225/529
$150 + 100$ answered 300 (differs on a tested digit)	0	225/1058
$150 + 100$ answered correctly	1/46	383/1058
errors on the practiced $100 + 100$	0	225/782
the same, after also studying $100 + 300$	0	3375/5842

ACT-R: confusion is graded, nothing is retrieved for certain, and new practice interferes with old.

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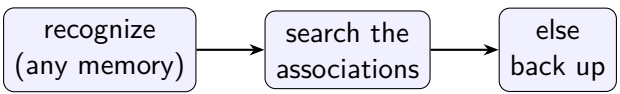
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ACT-R: confusion is graded, nothing is retrieved for certain, and new practice interferes with old.

- EPAM's zeros are theorems (no interference; near misses bounded by the tested digits). ACT-R's departures are proved on this instance and explained by laws it lacks.

The composition, extracted

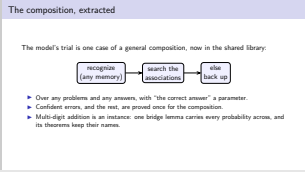
The model's trial is one case of a general composition, now in the shared library:



- Over any problems and any answers, with "the correct answer" a parameter.
- Confident errors, and the rest, are proved once for the composition.
- Multi-digit addition is an instance: one bridge lemma carries every probability across, and its theorems keep their names.

Recognition, retrieval and confident errors

The composition, extracted



- The multi-digit definitions were left untouched, so the Lean export used by the simulation and the paper is byte-identical.

A second task: naming people

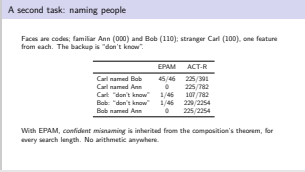
Faces are codes; familiar Ann (000) and Bob (110); stranger Carl (100), one feature from each. The backup is "don't know".

	EPAM	ACT-R
Carl named Bob	45/46	225/391
Carl named Ann	0	225/782
Carl: "don't know"	1/46	107/782
Bob: "don't know"	1/46	229/2254
Bob named Ann	0	225/2254

With EPAM, *confident misnaming* is inherited from the composition's theorem, for every search length. No arithmetic anywhere.

Recognition, retrieval and confident errors

A second task: naming people



- With ACT-R, the theorem's hypothesis (Carl's face calls up Bob's for certain) fails, and so does its conclusion: Carl says "don't know" more often than Bob.

From laws: which memories give confident errors?

Theorem. Any memory with three laws (*determinacy*, *certainty*, *stability*) produces confident errors whenever a stranger can be taken for a studied item with sharp associations.

- In the matrix of memory families, only discrimination (EPAM) has all three.
- Face naming's confident misnaming is derived this way, from EPAM's matrix cells plus one computed fact (Carl can be taken for Bob).
- The evidence judges that people lack certainty. So explaining confident errors in people needs a graded version: these laws with high probability, not probability 1.

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- This is where the composition and the law matrix meet: the matrix now says which families produce the phenomenon.
- The graded theorem is open.

Limits, and next

- **No timing.** "Retrieved, with no extra backup use" is what is proved; "fast" needs a timing model.
- **One trial at a time.** Learning across trials (feedback, discrimination, strengthening) makes memory states random; the predicted sequence (confident error, then column work, then fluent retrieval) is the next target.
- **Formal fidelity.** ACT-R here is one restricted formalization; checking it against PyACTUp, the community implementation, is planned.
- **A graded confident-errors theorem**, for memories without certainty.

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Limits, and next

- The blind simulation found that the learning sequence depends on what a newly discriminated problem starts with: no associations gives the predicted sequence in 40 of 48 cases; copying the neighbour's associations can trap the error.

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